Module 5: PPG and rPPG

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Laboratory of Nearable Technologies for Health Data Science
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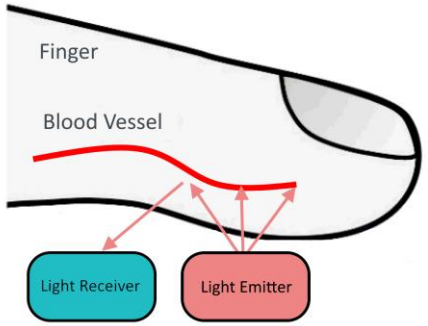
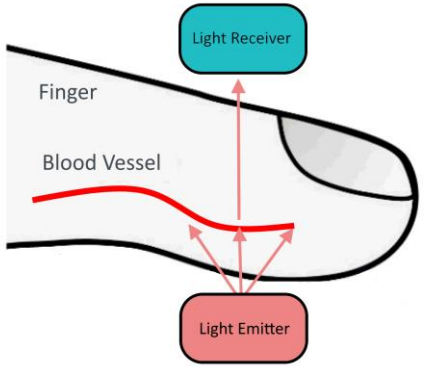
PPG

PhotoPlethysmoGraphy-PPG working principals



Using reflected light

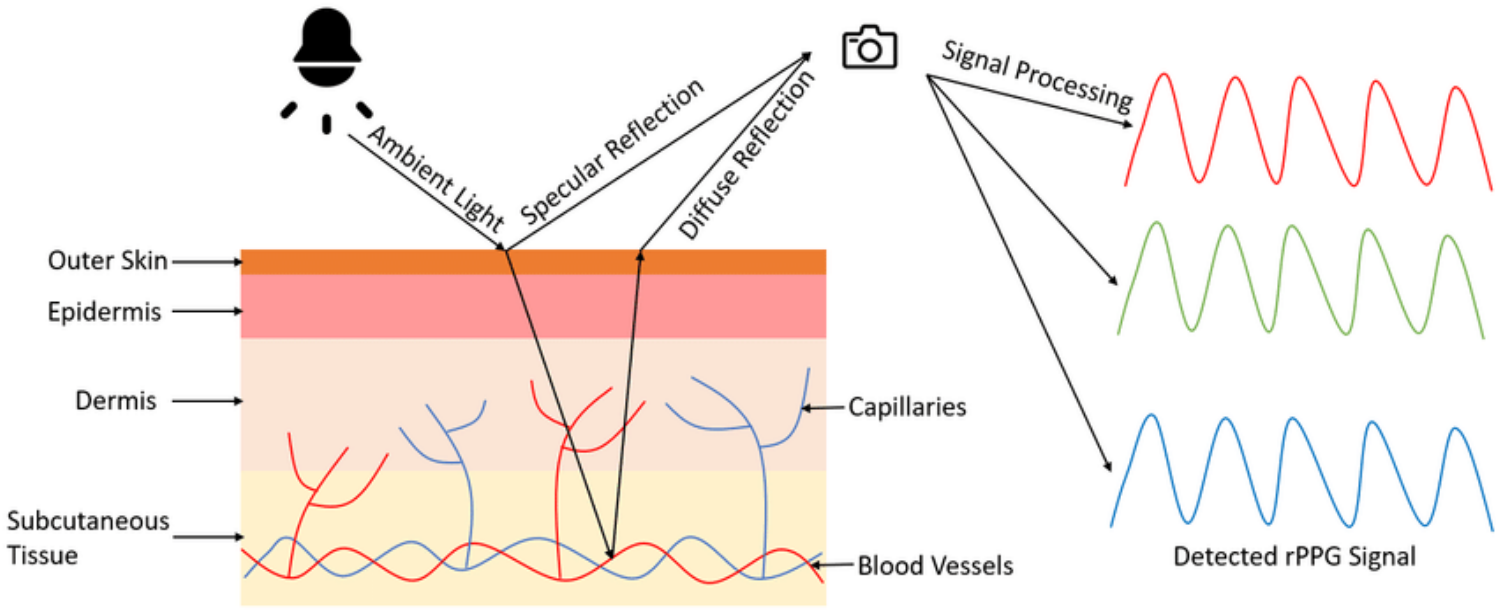


Using transmitted light

- When bloods flows into vessel it changes the reflective and transmitted coefficients
- By studying the dynamic of transmitted or reflected light intensity heart related features could be extracted: heart rate (HR), heart rate variability (HRV), respiratory rate etc.
- Using two different light emitters (red and infrared light) oxigenation ratio could be estimated (pulsoximeter)

rPPG

Remote PhotoPlethysmoGraphy-rPPG working principals

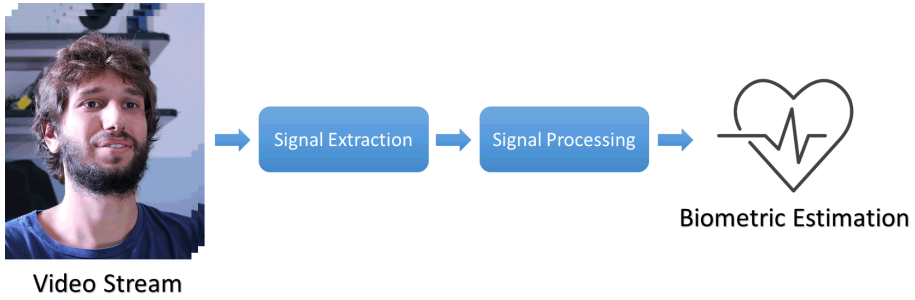


Using reflected light

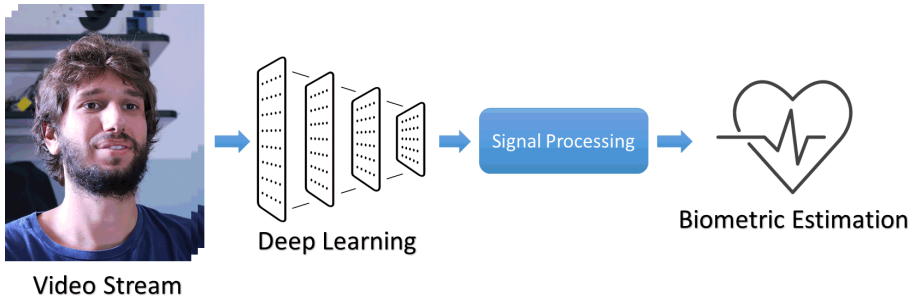
- Same principle as before
- Ambient or controlled light could be used
- Received light is a mix between: specular reflection (no PPG signal), diffusive reflection (PPG signal!), environment light and noise
- Signal is much more noisy!

rPPG

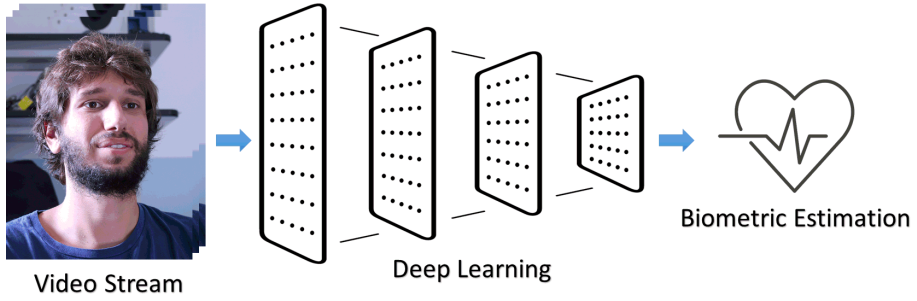
The use of deep learning in rPPG pipeline



Traditional approach
Signal processing could be used both for signal extraction (face detection) + signal processing (filtering, PCA, ICA, FFT)



Mixed approach
Deep learning is just used for signal extraction (skin segmentation)



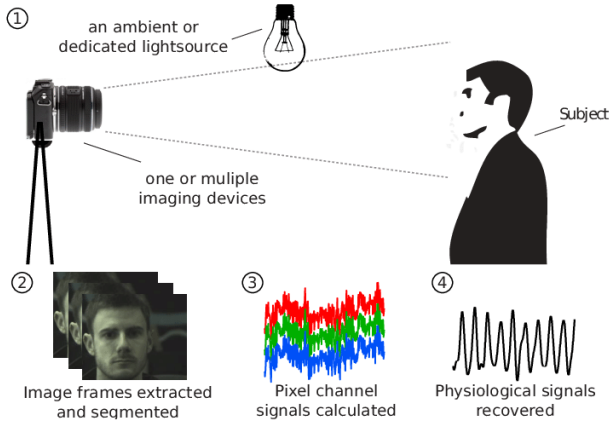
End-to-end deep learning
An end-to-end deep learning pipeline is used to directly map video stream to estimation of HR, HRV, etc.

Projects Proposals - rPPG

Heart Rate Monitoring

Wearables

PPG groundtruth



Nearables

Remote
photoplethysmography

The main goals

❖ Acquisition of raw data;
❖ Machine learning/Deep learning models: Extract biomedical information from wearable and nearable sensors;
❖ Graphical User Interface to visualize the data and the extracted biomedical data.

Today’s Agenda

Wearable PPG

- Installation/Setup
- Exercise: HR estimation

PPG using camera (rPPG)

- Exercise: simple rPPG with webcam
- Exercise: HR estimation using pyVHR

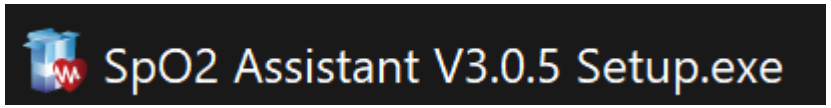
Pulse oximeter

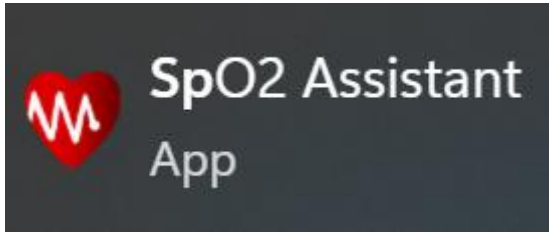
CMS50D Pulse Oximeter

- SpO2 (saturation)
  - Measuring range: 0%-100% Accuracy: ±2%
- Pulse rate (HR)
  - Measuring range: 30bpm-250bpm Accuracy: ±2 bpm
- Optical Sensor
  - Red light (wavelength is 660nm, 6.65mW)
  - Infrared (wavelength is 880nm, 6.75mW)
- USB connection
- Output: HR, SpO2, pulse wave

Pulse oximeter

Installation

- Download software from http://www.dlsoftw.com code:01YAG014
- Run 

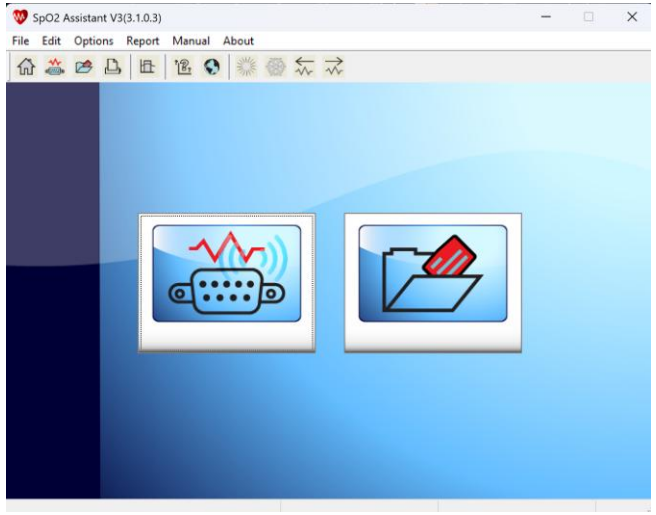
Unfortunatly MAC OS is not supported
- Connect Device with USB
- Turn on device and insert finger
- 

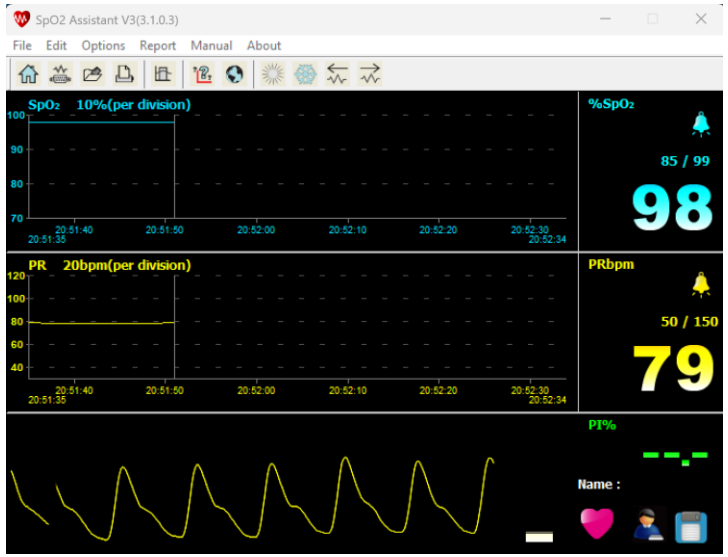
Pulse oximeter

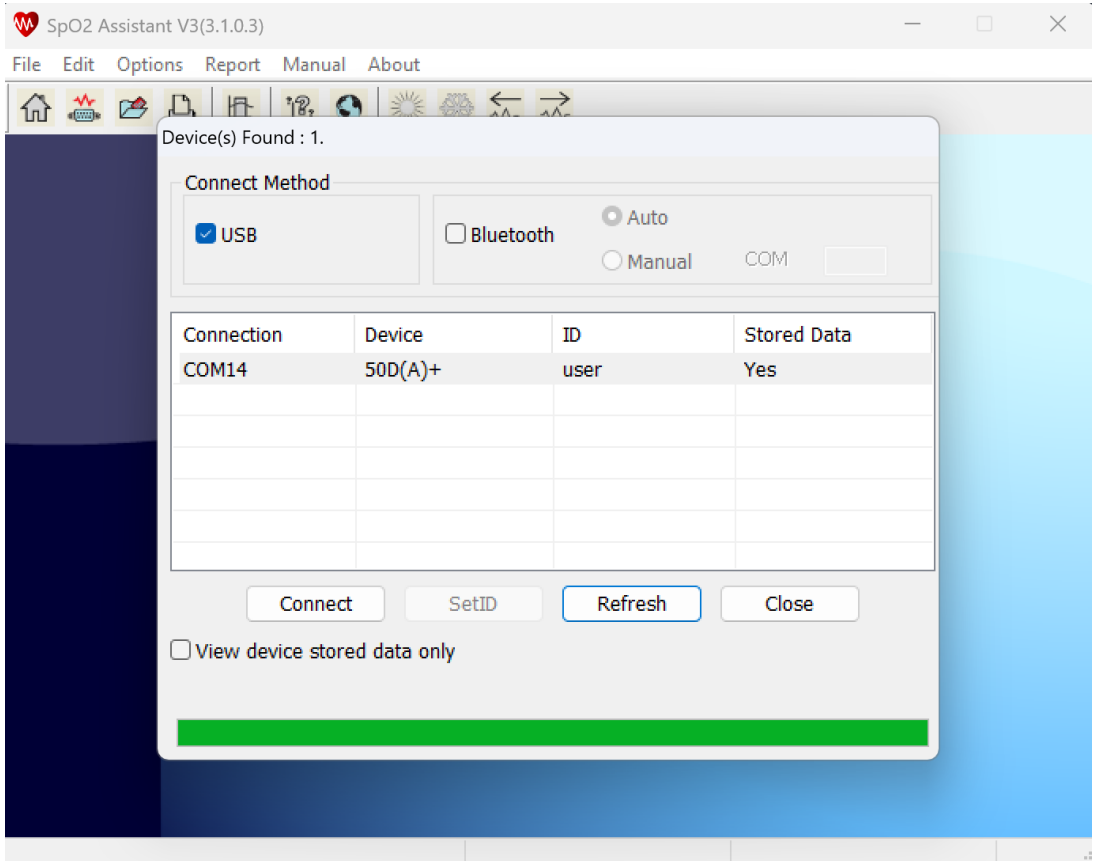
Installation

- Connect to device

- Run program







Pulse oximeter

Install pyserial and hidapi library

pip install pyserial hidapi

Download cms50d.py (it’s in the repo)

Run ppg.py to test everything is working fine

you might need to change COM port/HID (copy it from previous list)

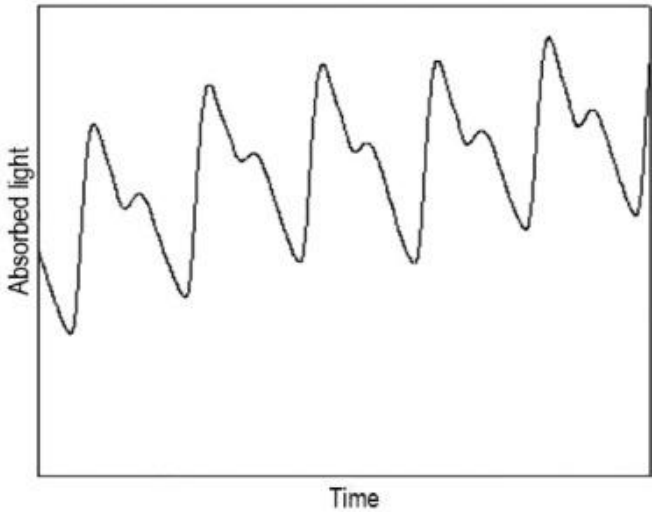
```
from cms50d import CMS50D


monitor = CMS50D(port="COM16")  # Replace with your actual COM
monitor.connect()
monitor.start_live_acquisition()
```

Exercise – Pulse oximeter

Modify the test script you used previously (ppg.py) to extract HR from the received pulse wave.

In particular

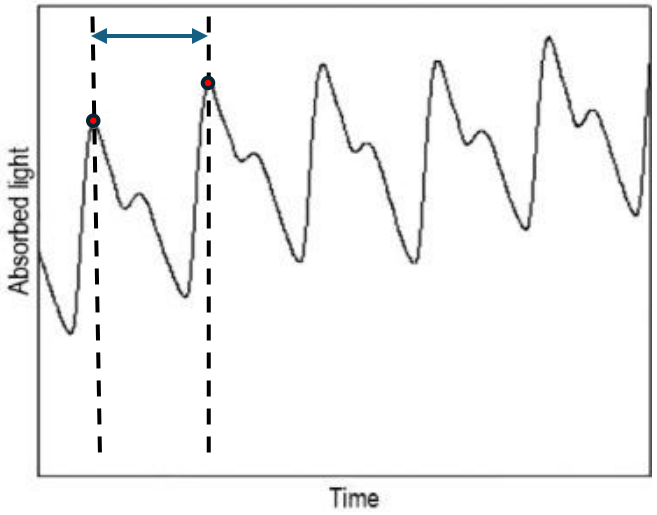


- Estimate HR using FFT (you can take the code from the first lab on signal processing)
- Show on graph estimated HR and HR data coming from the device

Exercise – Pulse oximeter

Modify the test script you used previously (ppg.py) to extract HR from the received pulse wave.

In particular

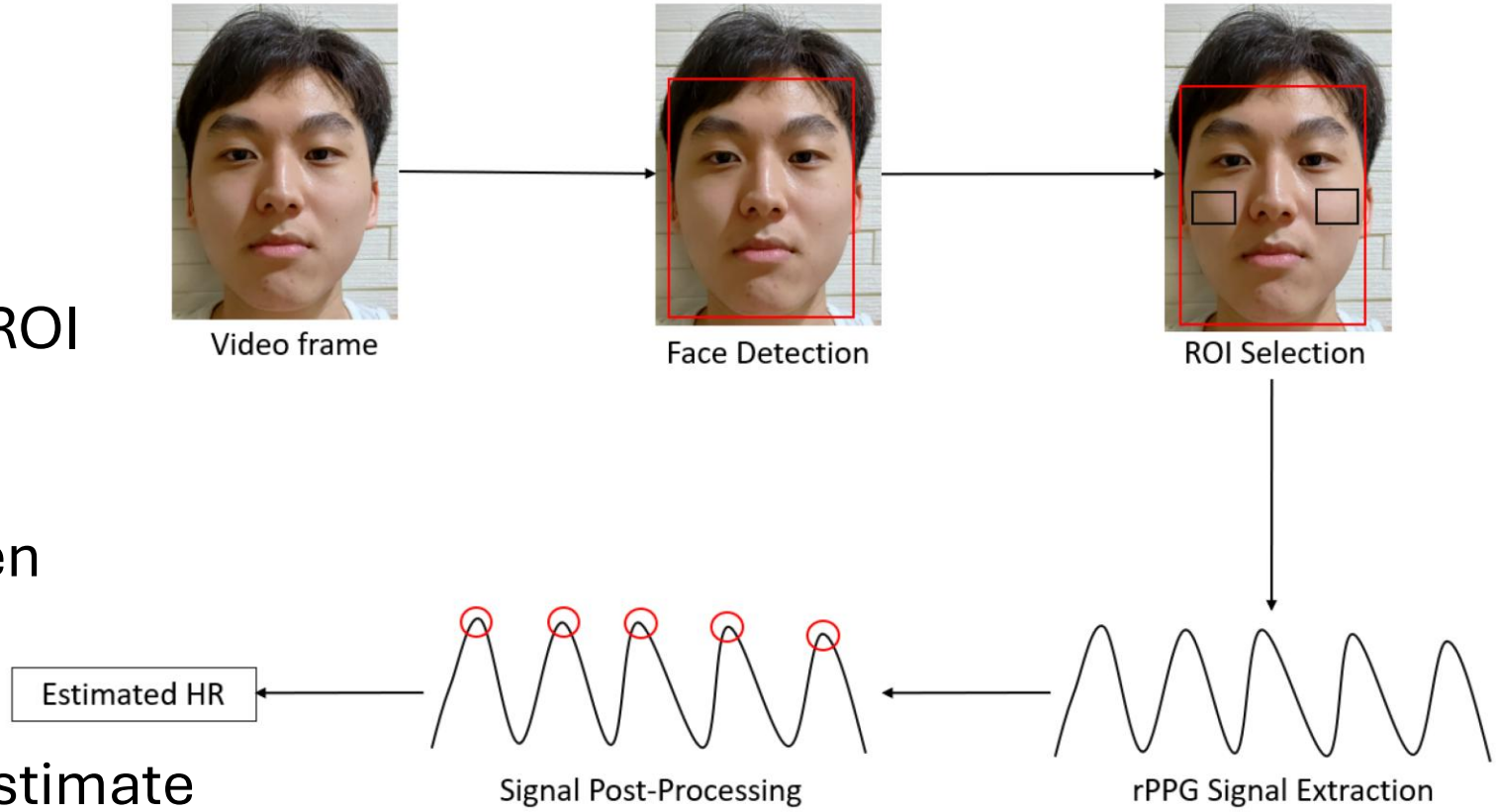


- Estimate HR using FFT (you can take the code from the session on signal processing)
- Estimate HR using peak detection (you can use find_peaks in scipy.signal)
- Show on graph estimated HR and HR data coming from the device

rPPG

Naïve rPPG

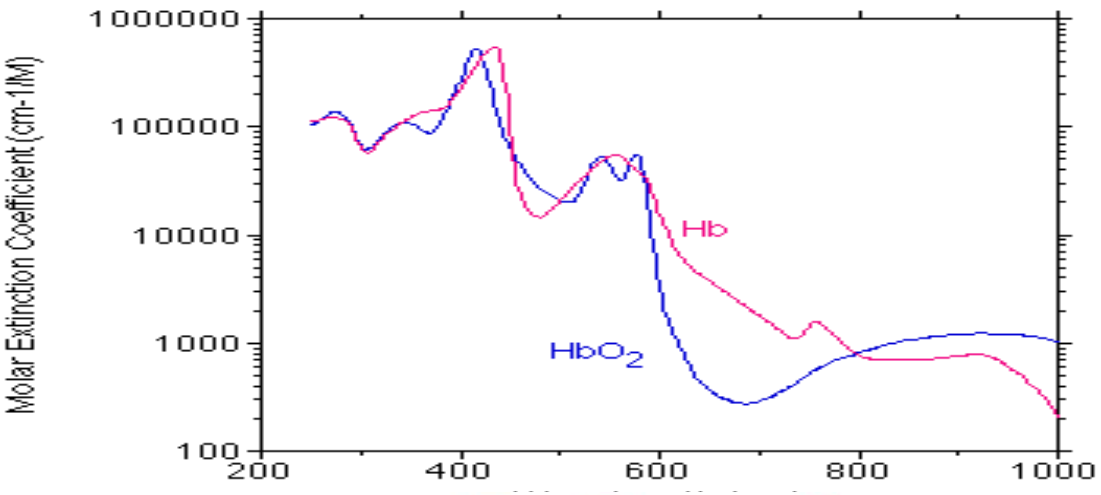
- Use face detection to find face bounding box
- Use fixed proportion to select ROI (usually forehead or cheeks)
- Compute average value of green channel of pixels inside ROI
- Use peak detection or FFT to estimate HR

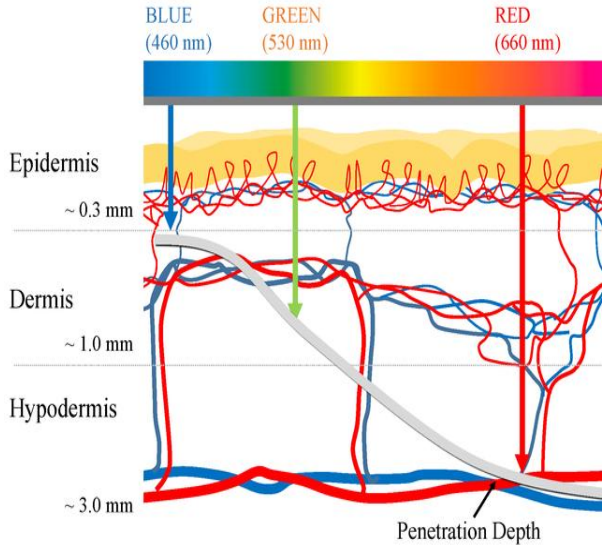


rPPG

Why the green channel?

- Hemoglobin strongly absorbs **green light (~520–580 nm)**
- Green light samples the **superficial capillaries**, where pulsation is most visible
- RGB cameras have 2× more green pixels (Bayer pattern)





Exercise - rPPG

Implement Naïve rPPG

- Implement face detection on live webcam stream (you can use code from image processing session)

- Select the ROI (forehead region) from face bounding box
  - forehead_x = face_x + 0.3*face_w
  - forehead_w = 0.4*face_w
  - forehead_y = face_y+0.08* face_h
  - forehead_h = 0.18*face_h

- Extract signal by concatenating average of green channel inside ROI

- Estimate HR using FFT

Exercise - rPPG

Implement Naïve rPPG

- Add bandpass filter (between 40 bpm and 220 bpm) (you can use code from signal processing session)
- Estimate HR using peak detection
- Add ROI stabilization

  curr_roi = alpha * curr_roi + (1-alpha)*prev_roi

rPPG – advance methods

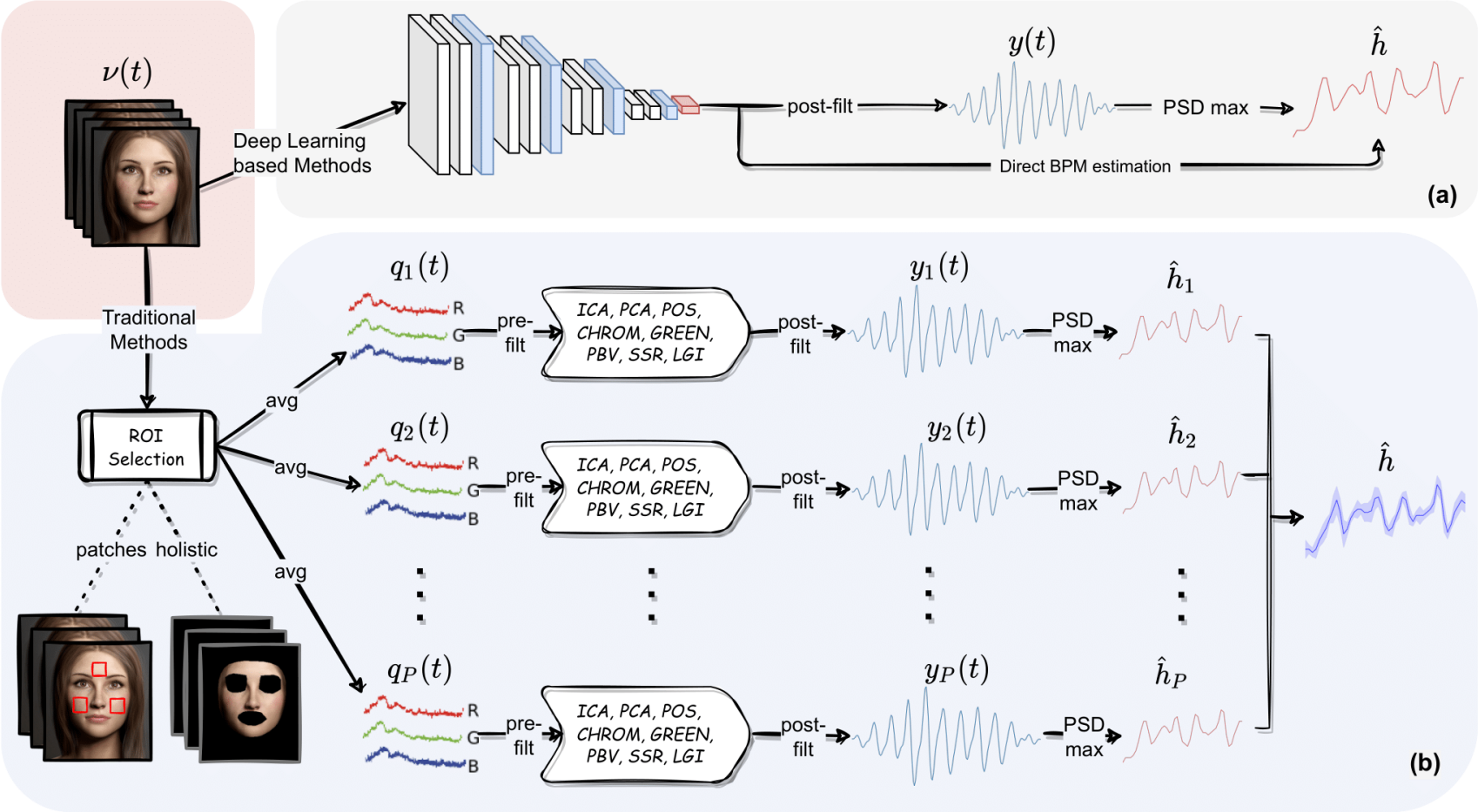
Main features of advance rPPG methods

- Deep learning for face detection and facial landmarks or skin segmentation
- Multiple patches to extract multiple signals
- PCA/ICA for fusing RGB signals
- Motion compensation
- End to end deep learning methods

rPPG – advance methods

pyVHR

Python library with implementation of main SOTA rPPG algorithms



Alternative: rPPG-Toolbox